

Maximizing profits and expanding into new sources of revenue for a software company

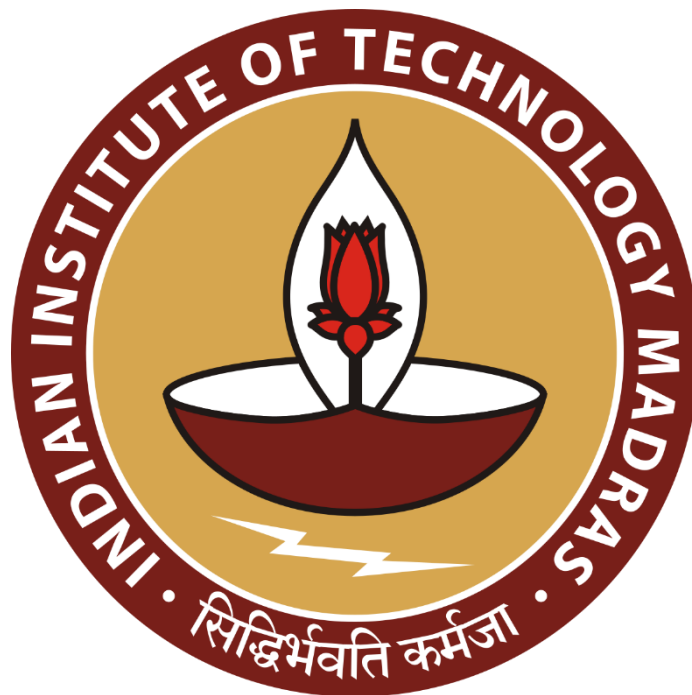
A Final report for the BDM capstone Project

Submitted by

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1. Executive Summary

This report looks into BLISS Infotech PVT LTD, a medium to large scale company that designs software specifically for life insurance agents and professionals in the financial sector. They are headquartered in New Delhi and have dealers and resellers spread out in most parts of the country.

The company is unable to maximize its profits due to poor relationship with dealers, mismanagement of intellectual property by the resellers and a lack of focus upon newer sources of revenue such as rural areas. Also, they do not have a support team which can look into the problems the users of the app may face and fix them on time.

This report has analysed the data that has been provided to us by the company and has provided recommendations and solutions for the company to implement in order to solve the problems it faces. These solutions include a survey of the different sources for marketing, following the practice of cross-training among the employees in order to increase their scope of work. Also, mentioned in this report is the need of a credit limit set by the company to the dealers and resellers, in order to keep a system of checks and balances between the company and its B2B partners.

This report contains details about all the methods and processes of analysis, the results and findings that have been concluded from the data and solutions that the company can choose to implement to solve their problems.

2. Process and Method of analysis

The key to achieve a fruitful reward from the data analysis is by breaking it down into different steps such as-

- Approaching and establishing a relationship with a business in order to understand their shortcomings and receive data so as to solve them
- Properly defining and understanding the problems
- Collecting the data relevant to the problems
- Cleaning and transforming the data in order to facilitate proper analysis
- Using the right tools for performing the analysis
- Obtaining and analysing the results so as to achieve strong and reliable solutions to the problems

1. The story of data acquisition

The method of approaching a business asking for their data is a process that involves talking to the business executives and forming a relationship of trust and credibility, as without it they may not wish to share their data. Before this company, I had tried four times to convince a business to collaborate with me on this report and share their data. Going in and asking for their data without actually establishing any trust resulted in a lot of reluctance and rejections from the business side.

At last, I met the founder and managing director of the company through a mutual friend and was able to convince him to share data regarding the operations of the company. After a quick background check on me from the board of the company, I was handed the data regarding the sales, marketing expenditure and employee structure of the company of March, April and May of Financial Year 2025.

2. Collection of required data

For collecting and storing the data generated by the business the company has a Management Information System or MIS which stores the data of the key aspects of the company since its inception. The company was kind enough to give me a virtual tour of their MIS and let me see for myself how their data was stored. The software on the click of a button generates a report in excel containing data of the timeline chosen by the user.

Link for the working dataset with graphs and analysis-

https://drive.google.com/drive/folders/1bEgVsmTdC4dNZe4_a4ZG1W3gsYI3nPF4?usp=drive_link

3. Processing and Cleaning of the data along with method of analysis

The data given to us by the company is in the raw format and needs to be cleaned and checked for any errors that may exist in it. After this step the data is converted to excel and organized so that the different spreadsheets in an excel file represent the different months of the respective data

Here is the list of all the datasets and the method of cleaning and analysis performed on them-

1. Sales Data

This dataset consists of all the business partners the company is working with to sell its software to the end customers. The raw data consists of the following columns-

- Customer Name- The name of the client.
- Mobile- The mobile number of the respective dealer.
- Email- The email id of the given dealer.
- Group- The company assigns the dealers groups based on their performance.
- Division- A geographical area wherein the dealer operates.
- Branch- A sub group of Division.
- Location- In order to classify the dealers into Urban or Rural based on their Division and Branch.
- Revenue- The amount of money in INR generated through selling the software by the respective dealer in a given month.

In the process of cleaning, any unwanted columns are removed and all the rows are checked for any inconsistencies. For analysis only three columns are considered. They are Customer Name, Location and Revenue.

The excel file containing the clean data has three spreadsheets. Each one of these spreadsheets contains the different dealers and resellers associated with the company for the given month.

In the month of March 2025, the company had 134 business partners. Similarly, for April, there were 123 dealers and for the month of May there were 113 partners. Also, the sort and filters feature of excel was used in order to fix some errors such as the design of the cells of the raw data as some of the cells had combined with their neighbouring cells and thus, were causing some trouble in the making of the graphs.

Table 1: Containing month-wise revenue

Month	Revenue
March	18044028
April	17025605
May	15624405

Key terminologies and their mathematical breakdown-

Total revenue = Sum of all revenue across all dealers

Average revenue of a dealer = Total revenue / No of dealers for the month

Growth rate of total revenue month on month-

$GR = \{(\text{total revenue of current month} - \text{total revenue of last month}) / \text{total revenue of last month}\} * 100$

Average growth rate across a given number of months-

$GRM = \text{Sum of GR of all months} / (\text{Total number of months} - 1)$

Percentage of sales in rural areas-

$PR(\text{Rural}) = (\text{Total revenue from rural areas} / \text{Total revenue}) * 100$

Percentage of sales in urban areas-

$PR(\text{Urban}) = (\text{Total revenue from urban areas} / \text{Total revenue}) * 100$

For analysis, Graphs and tables will be used to visualize the data on a monthly basis and understand trends and patterns to suggest solutions. Quantitative methods such as Skewness and Kurtosis are used to analyse the data and gain insights. The data will be binned into Rural and Urban in order to better understand the geographic division of sales.

2. Marketing Data

This dataset contains the marketing expenditure of the company in the months of March, April and May of 2025. This data consists of the columns-

- M_techniques – It is the list of the different marketing methods employed by the company
- Cost – It is the amount in INR spent on the respective marketing method for the given month

The company employs six methods of marketing and spends approximately two to three percent of its monthly revenue on marketing.

Further analysis of this data will be achieved through graphs and tables on Excel in the Results and Findings section. Skewness and Kurtosis are the quantitative methods used in the analysis of the data.

The results will be focused on maximizing the impact of the money spent on marketing.

3. Organization structure

This dataset contains the list of employees working for the company and the department in which they work. The columns are-

- Name - It consists of the name of the given employee
- Dept – Contains the department of the company in which the employee works

Currently there are 17 employees in the company working in departments such as Administration, Management, Development, Graphic Designing, etc.

Due to the small size of the marketing and employee data, the process of cleaning can be easily done through a visual inspection.

Further analysis on the dataset will be done through graphs in order to gain insights and suggest better collective resource management within the company.

3. Results and Findings

A. Graphs and visualization of the Sales data

Table 2: Containing description of the data regarding Sales

Statistic	March	April	May
Mean	135,669.38	139,554.14	139,503.61
Standard Error	4,921.00	5,556.53	5,803.13
Median	137,362.91	144,230.93	145,768.17
Mode	27,824.59	25,016.02	24,522.05
Standard Deviation	56,751.71	61,373.83	61,414.59
Maximum	290,694.39	291,588.76	295,210.43
Minimum	27,824.59	25,016.02	24,522.05
Count	133	122	112
Sum	18,044,027.65	17,025,604.82	15,624,404.78
Skewness	0.13	0.16	0.13
Kurtosis	-0.04	-0.08	-0.08
Range	262,869.80	266,572.74	270,688.38

Insights gained from the table are-

- Mean sales rose from INR 135,669.38 (March) to INR 139,554.14 (April), then stayed roughly the same in May (INR 139,503.61).
- While the total sales dropped, the average per dealer improved slightly, suggesting fewer dealers but relatively stable high-value transactions.
- There is a mild left skewness as indicated by the values (0.13~0.16). This means that there may be some lower-end outliers pulling the mean slightly down.
- The value for kurtosis is slightly negative (-0.04~-0.08). This means that the distribution is platykurtic, basically flatter than normal, with fewer extreme values.
- Std dev is greater than INR 56,000 across all months, showing wide variability in sales among dealers.

- Range is increasing each month from about INR 262K in March to about INR 271K in May, suggesting widening gap between top and bottom performers.
- Mode is much lower than the mean and median in all months from about INR 25K to 28K, implying a large cluster of low-performing resellers, and a few high outliers raising the average. Coincidentally, this happens to be one of the problems that the company says it suffers from and is now clearly represented by the data upon analysis.
- The number of dealers the company is associated with is also on a decline, from 132 in March, to 122 in April and finally 112 in May. This is an alarming statistic as it suggests that the company is losing dealers and hence making less revenue month by month.

Graph 1



This graph shows the trend of revenue from the month of March to the month of May. The numbers of the revenue are-

- Revenue of March - INR 18044028
- Revenue of April – INR 17025605
- Revenue of May – INR 15624405

This data obtained from the chart will help us in figuring out the percentage of revenue obtained based on location. The calculation for these will be done in the next three graphs.

The general insight from this graph is that the company has had a decline in its month on month revenue which is evident due to the downward slope of the graph. This, the company suggests, is due to the issue of cross-selling and poor dealer management resulting in a reduction of number of dealers. Solutions for this will be mentioned in the section 'Implementation of Results and Findings'.

Graph 2



This graph divides the revenue of the month of march into urban and rural. The statistics for this graph are-

- Revenue generated from urban areas = INR 17476380.96
- Revenue generated from rural areas = INR 567646.69

From these numbers we can figure out the percentage of sales in rural and urban areas using the formula- $(\text{Revenue generated from a specific location} / \text{Total Revenue}) * 100$

The numbers for these are-

- Percentage revenue generated from urban areas = $(17476380.96 / 18044027.7) * 100$
= 96.854 %

- Percentage revenue generated from rural areas = $(567646.69 / 18044027.7) * 100$
= 3.145 %

The same calculations have been done below for Graph 3 and Graph 4

Graph 3



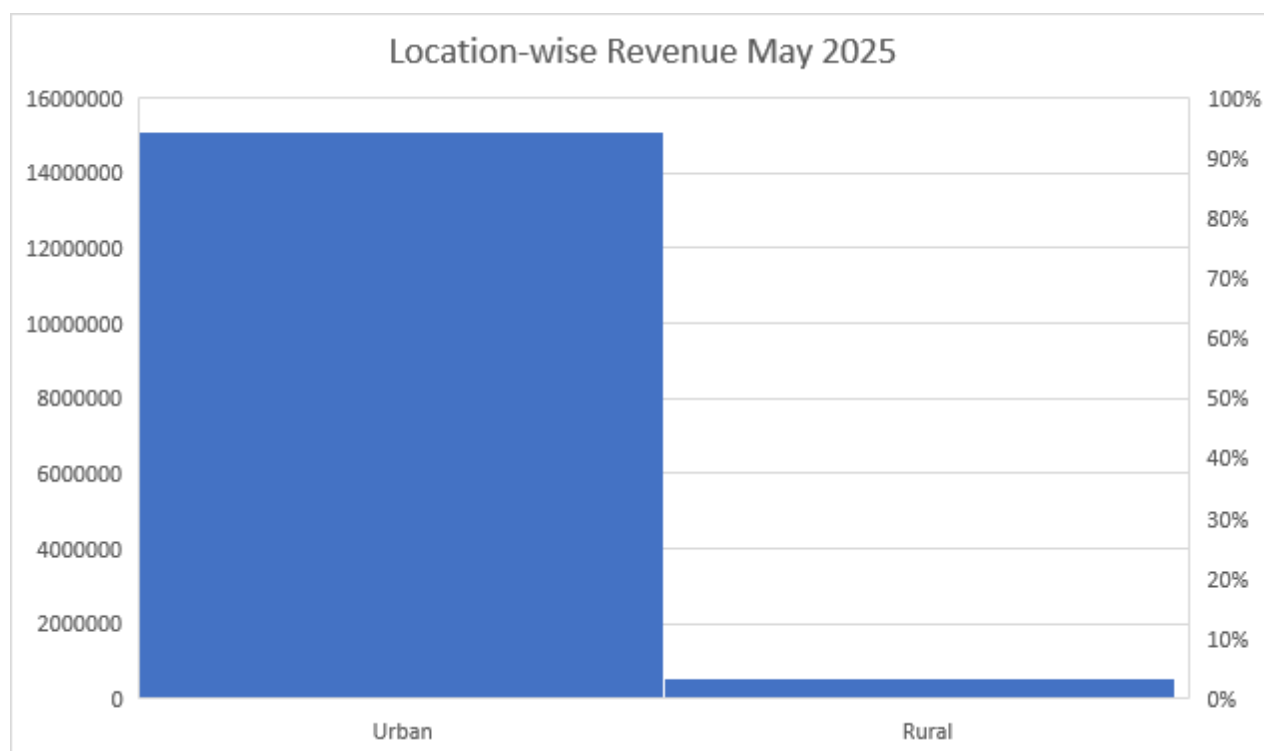
The statistics for this graph are-

- Revenue generated from urban areas = INR 16463260.4
- Revenue generated from rural areas = INR 562344.42

The Percentages for these are-

- Percentage revenue generated from urban areas = $(16463260.4 / 17025604.8) * 100$
= 96.697 %
- Percentage revenue generated from rural areas = $(562344.42 / 17025604.8) * 100$
= 3.302 %

Graph 4



The statistics for this graph are-

- Revenue generated from urban areas = INR 15087724.18
- Revenue generated from rural areas = INR 536680.6

The Percentages for these are-

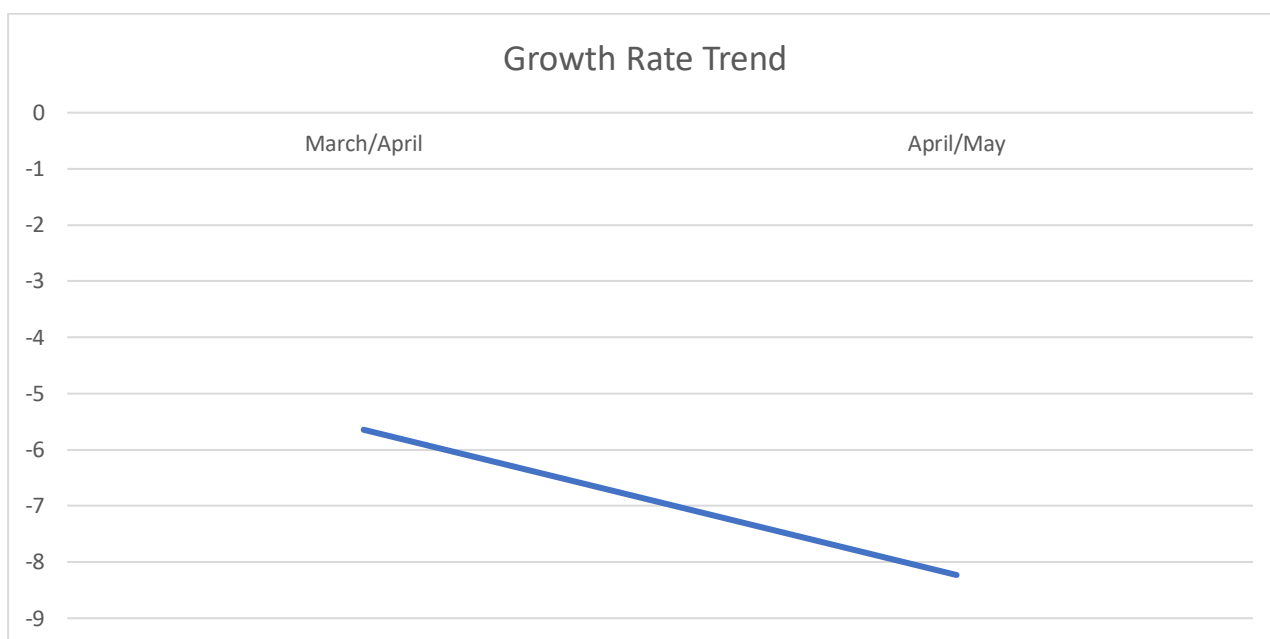
- Percentage revenue generated from urban areas = $(15087724.18 / 15624404.8) * 100$
= 96.565 %
- Percentage revenue generated from rural areas = $(536680.6 / 15624404.8) * 100$
= 3.343 %

Insights from graph 2, graph 3 and graph 4 –

- Approx. 96 percent of the revenue is from urban areas and approx. 3 percent from the rural areas which suggests an overdependence on urban areas.
- This can be a dangerous condition as, if the company starts to underperform in the urban sector or gets beaten by a rival company, it will lose the only source of revenue it has.

Solutions to curb these problems will be mentioned in the next section.

Graph 5



This graph shows the growth rate of revenue month after month. The numbers on this graph are calculated using the formula-

$$GR = \{(\text{total revenue of current month} - \text{total revenue of last month}) / \text{total revenue of last month}\} * 100$$

The numbers and their calculations are-

- Growth Rate (March to April) = $\{(17025605 - 18044028) / 18044028\} * 100$
= -5.644 %
- Growth Rate (April to May) = $\{(15624405 - 17025605) / 17025605\} * 100$
= -8.229 %

The general insight from this graph is that not only is the revenue of the company decreasing month on month as indicated by the negative values of growth rate and graph 1, but also that the rate at which the company is losing revenue is also increasing from 5.644 percent to 8.299 percent. This means that the company may be on a runaway, which cannot be calculated due to a lack of information on the company's total expenditure and that it is underperforming even in the urban sector which commands the majority of their revenue.

Thus, they are in an urgent need for solutions to fix these problems. Otherwise they may run into deep and chronic financial trouble.

Also, now we can calculate the average growth rate across the months using the formula-

Average GR = Sum of GR of all months / (Total number of months – 1)

$$= \{(-5.644) + (-8.299)\} / (3 - 1)$$

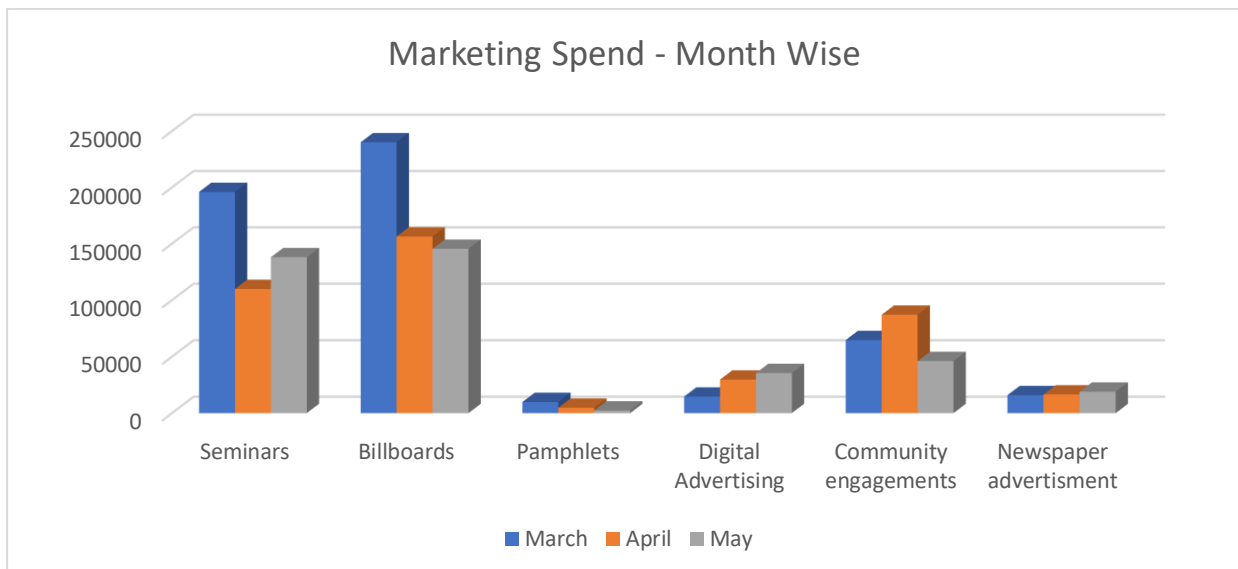
$$= -13.943/2$$

$$= -6.971$$

This means that the company's revenue has been declining at an average rate of 6.971 percent over the three months of sales data that we have been provided.

B. Graphs and visualization of the Marketing data

Graph 6



The Insights gained from this graph are-

- March saw the highest overall spending.
- Traditional media such as Seminars and Billboards peaked early and then declined.
- Digital spending rose along the months thus, indicating a shift toward online marketing.
- Pamphlets and community engagement in May might reflect budget reallocation or seasonal completion of marketing campaigns organized by the company.
- Newspaper remained the most stable channel, likely used for ongoing brand reinforcement.

Table 3: Contains a description of data regarding Marketing

Statistic	Billboards	Community engagements	Digital Advertising	Newspaper advertisement	Pamphlets	Seminars	Total
Mean	180,566	65,866	26,550	17,100	5,620	147,866	443,570
Std. Error	29,882	11,852	6,201	916	2,207	25,334	48,874
Median	156,300	64,600	29,650	16,500	4,760	137,940	404,010
Mode	145,400	46,000	14,600	15,900	2,300	109,800	385,940
Std. Dev.	51,758	20,529	10,740	1,587	3,823	43,880	84,652
Max	240,000	87,000	35,400	18,900	9,800	195,860	540,760
Min	145,400	46,000	14,600	15,900	2,300	109,800	385,940
Sum	541,700	197,600	79,650	51,300	16,860	443,600	1,330,710
Range	94,600	41,000	20,800	3,000	7,500	86,060	154,820
Skewness	1.65	0.28	-1.19	1.46	0.96	0.97	1.64
Kurtosis	-1.50	-1.50	-1.50	-1.50	-1.50	-1.50	-1.50

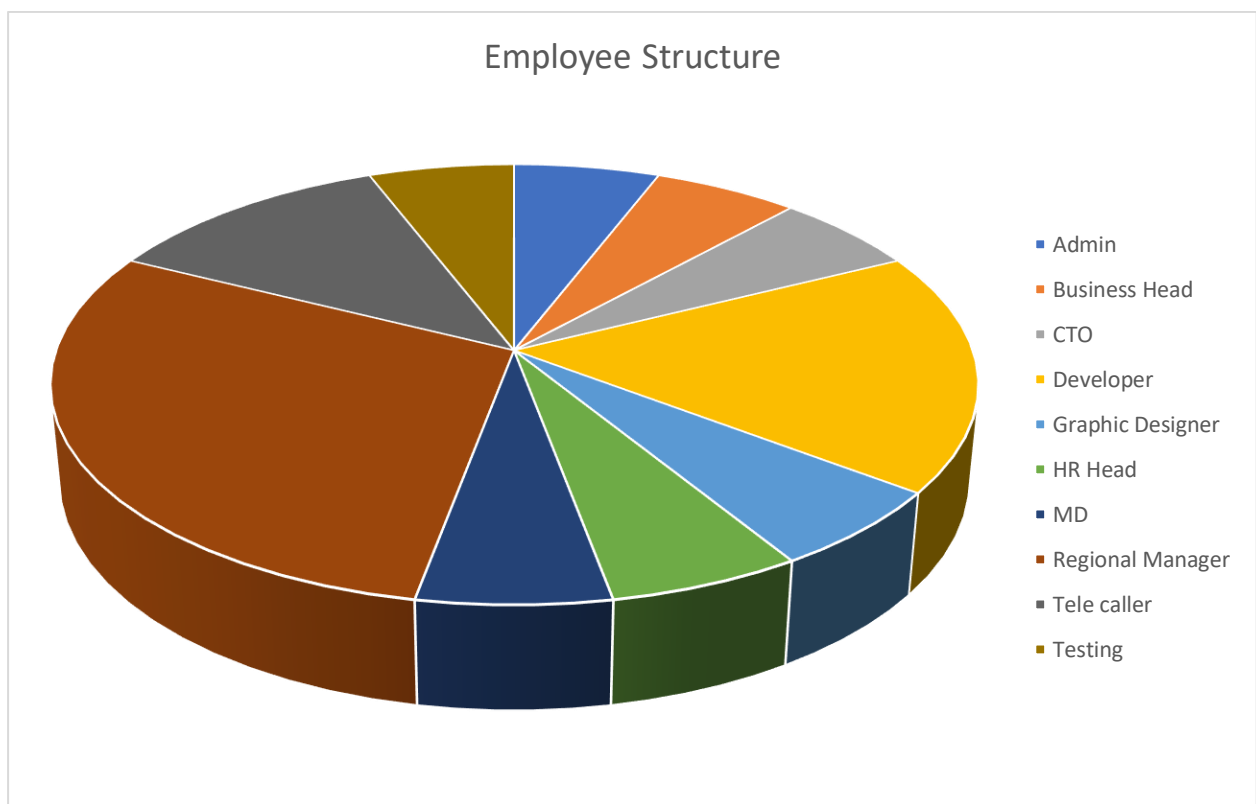
The Insights gained from the table are-

- Billboards (INR 180,566) and Seminars (INR 147,866) dominate the average spending. These account for over 73% of the Total Mean (INR 443,570).
- Pamphlets (INR 5,620) and Newspaper advertisement (INR 17,100) had the least investment.
- Billboards and seminars have the highest standard deviation, meaning that they can be considered as the most variable techniques, wherein the spending is very inconsistent.
- On the other hand, newspapers have the least standard deviation and hence can be considered as the most stable form of marketing.
- Seminars with 0.97, Billboards with 1.65, Pamphlets with 0.96, and Newspaper Advertisement with 1.46, all show positive skewness, meaning: Most expenditure values are clustered at the lower end, with a few larger values pulling the mean higher.

- Community Engagements with 0.28, shows mild positive skewness, meaning spending is fairly evenly distributed but still slightly weighted toward lower values.
- Digital Advertising with -1.19, shows a moderate negative skew, implying: Most values are clustered at the higher end, but a low value (March: 14,600) is pulling the mean down.
- All techniques have a kurtosis of -1.50, which is platykurtic, this means that the distributions are flatter than normal, with lighter tails, this implies fewer extreme outliers, suggesting relatively controlled spending behaviour.
- Pamphlets and Digital Advertising offer low-cost options with less variability, making them more predictable and controllable.
- The Total marketing spend shows a very high range (INR 154,820) and high std. deviation (INR 84,652). This indicates a substantial fluctuation in overall monthly budget.

C. Graphs and visualization of Employee structure

Graph 7



This graph shows us the breakdown of the employees and the department they work under.

The numbers obtained from the graph are-

Department	Count
Regional managers	5
Tele Caller	2
Developer	3
CTO	1
Testing	1
Graphic Designer	1
Admin	1
HR Head	1
Business Head	1
Managing Director	1

The insights from these numbers are-

- There is no employee in a dedicated support team which can solve any problems that occur after the product is launched.
- The number of people in the development and testing teams compared to the number of people in the management team is quite low for a company that specializes in developing software products.
- While it is necessary to have managers to handle the relationships with the dealers and resellers, it is also just as necessary to have a good Tech-team in order to make good products with consistency and keep providing new features in them.

4. Interpretation of Results and Findings

Here is the list of the key takeaways from the previous section-

1. Sales and Dealer Performance

- **Decline in Revenue:** The total revenue fell from INR 1.804 CR in March to INR 1.562 CR in May, with an average monthly decline of 6.97 percent. The rate of decline has been increasing each month, which indicates that there may be a growing number of operational issues.

- **Dealer Renewal Rate:** The number of active dealers kept on dropping month-on-month from 133 to 112. This indicates that there are problems regarding dealer engagement and retention.
- **Dominance in Urban Areas:** Approx. 96 percent of the revenue came from urban areas, while rural areas contributed to just over 3 percent. This pattern was seen to be consistent across all the three months and reflects one of the problems of the company i.e. an overdependence on urban markets. Hence, posing a risk to the overall revenue, if urban sales slump.
- **Distribution of revenue across the Dealers:** High std. dev. and low modal values of the data in the analysis table indicate a long-tailed distribution, meaning that there are a few high performing and many low-performing dealers. This justifies the company's claim about uneven dealer performance.

2. Marketing Expenditure Analysis

- **Concentration of Spending across the months:** Billboards and Seminars resulted for over 70 percent of the total spending in any given month, while Pamphlets and Newspaper ads had a very minimal investment made into them.
- **A Shift in Trend:** A shift towards digital marketing had been observed, with reduced spending on traditional channels along the three months of data given.
- **Stability and Volatility in Spending:** Newspaper ads were the most stable channel with a very low variability, while the spending on seminars and billboards were highly variable across the different months.
- **Potential to optimize:** Low-cost and low-variance options like pamphlets and digital ads offer a very cost-effective scalability, especially for underexplored rural markets, wherein the company is adamant to enter and diversify its source of revenue.

3. Organizational Structure

- **Small Team:** The company has just 17 employees in its core team. This includes only three people in the development team and one person in the testing team. These numbers are quite low for a tech-driven product firm.
- **Imbalance in Roles:** There are a lot of managers, for example, 5 regional managers, compared to core technical and support roles. This data supports the problem of the company not having enough support staff to handle any issues that may arise after the deployment of an app and hamper the experience of the end consumers.

List of Problems and their respective Solutions and Recommendations based on the results of the analysis of the data-

1. Mismanagement of software by dealers – Credit Limit

This is a major problem for the business and is known as ‘cross-selling’. In it a dealer, steals the intellectual property of the app and sells it as a product of their own making. In order to stop unauthorized sales of this manner, having a policy of a **credit limit** will help the company in reducing financial risk, ensure accountability among dealers and facilitate recovery of any dues or losses due to any partner-side mismanagement. This system should function in the following order-

- Evaluate each dealer and reseller based on their performance.
- Assign a credit limit to each dealer or reseller based on the parameters and discussions held with them. Ensure to take a security deposit from the dealer in advance and that the amount of the security deposit is greater than the assigned credit limit.
- Allow dealer and resellers to place orders only within the approved credit limit.
- Integrate the credit limit system into the Management Information System (MIS). This will allow the company to monitor the credit usage of a partner in real time.
- Send alerts to the partner when they approach near or exceed their credit limit.
- Block any further orders if the partner has gone over their credit limit.
- Provide the dealers with invoices with defined payment terms and due dates for the same.
- Track payments and update credit balance immediately after receiving the invoices through the MIS.
- Apply penalties for any delayed payments or defaults done by the partner.
- In case of any losses recover the dues through withholding security deposits, or legal action if needed.
- Review and revise limits periodically based on updated partner performance.

These are the steps to be followed in order to implement a proposed credit limit. Its basic job is to prevent any transaction regarding the software from the partners side to not go unnoticed by the company.

2. Drop in the number of B2B partners – Loyalty program

As shown in the analysis of the data the count of dealers dropped from 133 in March to 112 in May. In order to retain their partners, the company should devise a loyalty program wherein the partners are split into groups based on parameters such as performance and area. They are then given bonuses or rewards for performing well. The table below contains the proposed loyalty program.

Tier	Eligibility	Benefits
Silver	Monthly sales >50000 INR	1% Bonus on sales Basic Marketing Support Access to new product launches
Gold	Monthly sales >100000 INR And No defaults or pending payments	2.5% Bonus on sales Dedicated Regional Support Early access to leads Co-branded marketing materials
Platinum	Monthly sales >250000 INR For at least 3 months	5% Bonus on sales Exclusive Offers and Territory Protection Featured on website

3. Spend on Marketing

In order to optimize its expenditure on marketing the company must analyse all the people approaching the company and survey them regarding how they gained knowledge of the company's existence. This survey will yield critical results regarding the most effective source for marketing. Based on this data the company must assess its sources of marketing and spend accordingly. Ideally, in order to expand into rural areas, the company must focus on newspapers. On the other hand, Digital Marketing provides an urban-centric but cost-effective method of marketing as compared to billboards. Hence these changes should be implemented in the short run to see and analyse changes.

4. Employee Structure – Cross Training

In order to accommodate a support team, the process of cross training can be used.

Cross-training is the practice of training employees to perform any tasks or responsibilities outside their primary job role, enabling them to develop multiple skill sets across different functions or departments. From the details of the employee structure, we can take the Tele Callers, developers and testers and cross-train them in such a way that the tele-caller can be active on a support number and the developer can fix any bugs or errors in the application.

Also, the company must store data regarding any calls made to the support line number. The parameters of this data can be- Problems, Root Cause and Solution. This will help the company in learning from any mistakes made in its previous products and make better products and solutions in the future.